

# Bento Natura

*Assistant Professor · Columbia University*

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## RESEARCH INTERESTS

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Algorithms, optimization, and game theory. Specific interests include linear programming, interior point methods, convex optimization, combinatorial optimization, and circuit geometry of polytopes.

## POSITIONS

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**Assistant Professor in Industrial Engineering & Operations Research (IEOR)** *since 2024*  
Columbia University, *New York City, NY*

**Beerman Postdoctoral Fellow** *2022 – 2024*  
Georgia Institute of Technology, *Atlanta, GA*  
Joint affiliations with ISyE and the College of Computing.

**Postdoctoral Research Fellow** *Fall 2023*  
University of California, Berkeley, *Berkeley, CA*  
Long Term Participant, program on *Data Structures and Optimization for Fast Algorithms* at the Simons Institute.

**Postdoctoral Fellow** *Spring 2023*  
Brown University (ICERM), *Providence, RI*  
Program: *Discrete Optimization: Mathematics, Algorithms, and Computation*.

### *Industry & Other*

Visiting Associate, The Boston Consulting Group, *Munich, Germany* *2018*  
Junior Consultant, TNG Technology Consulting, *Munich, Germany* *2018*  
Research Assistant, Research Institute for Discrete Mathematics, *Bonn, Germany* *2014 – 2017*

## EDUCATION

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**PhD in Mathematics** *2018 – 2022*  
London School of Economics and Political Science, *London, UK*  
Supervisor: László Végh. Thesis: *Exact Linear Programming: Circuits, Curvature, and Diameter*.

**MSc in Mathematics** *2012 – 2017*  
University of Bonn, *Bonn, Germany*  
Thesis: *Algorithms for Routing and Buffer Insertion*. GPA: 4.0/4.0.

## HONORS & AWARDS

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### Richard Rado Prize

German Mathematical Society, *TU Berlin*

Biennial prize for the best dissertation in Discrete Mathematics.

*Oct 2024*

### Runner-Up for The Doctoral Award

The Operational Research Society (UK)

*Oct 2023*

### PhD Prize

Department of Mathematics, London School of Economics

*Jul 2023*

### Dodu Prize ♦ Best Talk of a Young Researcher

SMAI-MODE, *Université Paris-Saclay*

*Sep 2020*

## PUBLICATIONS

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In theoretical computer science, papers are typically first published at peer-reviewed conferences. Authors are ordered alphabetically.

### *Preprints*

- [P1] **B. Natura**. “Circuit Diameter of Polyhedra is Strongly Polynomial.” *In submission*, 2026. [ARXIV:2602.06958]

### *Journal Articles*

- [J1] X. Allamigeon, D. Dadush, G. Loho, **B. Natura**, L.A. Végh. “Interior Point Methods Are Not Worse Than Simplex.” *SIAM Journal on Computing*, 2025. [ARXIV:2206.08810]
- [J2] D. Dadush, Z.K. Koh, **B. Natura**, L.A. Végh. “On Circuit Diameter Bounds via Circuit Imbalances.” *Mathematical Programming*, 2024. [ARXIV:2111.07913]
- [J3] D. Dadush, S. Huiberts, **B. Natura**, L.A. Végh. “A Scaling-Invariant Algorithm for Linear Programming Whose Running Time Depends Only on the Constraint Matrix.” *Mathematical Programming*, 2023. [ARXIV:1912.06252]
- [J4] S. Jiang, **B. Natura**, O. Weinstein. “A Faster Interior-Point Method for Sum-of-Squares Optimization.” *Algorithmica*, 2023. [ARXIV:2202.08489]
- [J5] S. Daboul, S. Held, **B. Natura**, D. Rotter. “Global Interconnect Optimization.” *ACM Trans. Design Automation of Electronic Systems*, 2023.
- [J6] D. Dadush, Z.K. Koh, **B. Natura**, L.A. Végh. “An Accelerated Newton-Dinkelbach Method and its Application to Two Variables Per Inequality Systems.” *Mathematics of Operations Research*, 2022. [ARXIV:2004.08634]
- [J7] F. Ekbatani, **B. Natura**, L.A. Végh. “Circuit Imbalance Measures and Linear Programming.” *Surveys in Combinatorics*, 2022. [ARXIV:2108.03616]

## Conference Papers

- [C1] D. Dadush, H. Ma, **B. Natura**, L.A. Végh. “Trust Region Interior Point Methods: Optimal  $\ell_2$ - and Faster Wide-Neighborhood Path Following.” *STOC*, 2026.
- [C2] D. Dadush, Z.K. Koh, **B. Natura**, N. Olver, L.A. Végh. “A Strongly Polynomial Algorithm for Linear Programs with at Most Two Non-Zero Entries per Row or Column.” *STOC*, 2024. [\[PDF\]](#)
- [C3] X. Allamigeon, D. Dadush, G. Loho, **B. Natura**, L.A. Végh. “Interior Point Methods Are Not Worse Than Simplex.” *FOCS*, 2022. [\[ARXIV:2206.08810\]](#)
- [C4] D. Dadush, Z.K. Koh, **B. Natura**, L.A. Végh. “On Circuit Diameter Bounds via Circuit Imbalances.” *IPCO*, 2022. [\[ARXIV:2111.07913\]](#)
- [C5] S. Jiang, **B. Natura**, O. Weinstein. “A Faster Interior-Point Method for Sum-of-Squares Optimization.” *ICALP*, 2022. [\[ARXIV:2202.08489\]](#)
- [C6] **B. Natura**, M. Neuwohner, S. Weltge. “The Pareto Cover Problem.” *ESA*, 2022. [\[ARXIV:2202.08035\]](#)
- [C7] D. Dadush, Z.K. Koh, **B. Natura**, L.A. Végh. “An Accelerated Newton-Dinkelbach Method and its Application to Two Variables Per Inequality Systems.” *ESA*, 2021. [\[ARXIV:2004.08634\]](#)
- [C8] D. Dadush, S. Huiberts, **B. Natura**, L.A. Végh. “A Scaling-Invariant Algorithm for Linear Programming Whose Running Time Depends Only on the Constraint Matrix.” *STOC*, 2020. [\[ARXIV:1912.06252\]](#)
- [C9] D. Dadush, **B. Natura**, L.A. Végh. “Revisiting Tardos’s Framework for Linear Programming: Faster Exact Solutions using Approximate Solvers.” *FOCS*, 2020. [\[ARXIV:2009.04942\]](#)
- [C10] S. Daboul, S. Held, **B. Natura**, D. Rotter. “Global Interconnect Optimization.” *ICCAD*, 2019. [\[DOI\]](#)

## INVITED TALKS & PRESENTATIONS

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### *Selected Invited Lecture Series*

#### **14th Cargèse Workshop on Combinatorial Optimization**

*Sep 2025*

INRIA/CNRS, Corsica, France

Lecture series on *The Combinatorics of Interior Point Methods for Linear Programming*.

#### **Summer School on Linear Program Solvers**

*Jun 2025*

Georgia Institute of Technology, Atlanta, GA

Lecture series on *Strongly Polynomial IPM and Generalized Flows*.

*“A Unified Analytical Approach to Strongly Polynomial Algorithms for Structured Linear Programs”*

Oberseminar Discrete Optimization, University of Bonn, Germany

*Jul 2025*

### *Conference & Seminar Presentations*

*“Circuit Diameter of Polyhedra is Strongly Polynomial”*

Follow-Up Workshop: Discrete Optimization (Plenary), HIM, University of Bonn, Germany *Apr 2026*

*“A strongly polynomial algorithm for the minimum-cost generalized flow problem”*

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| Foundations of Data Science Seminar, Yale University, CT           | <i>Sep 2024</i> |
| Tutte Distinguished Lecture Series, University of Waterloo, Canada | <i>Sep 2024</i> |
| MOPTA 2024, Lehigh University, PA                                  | <i>Aug 2024</i> |
| ICERM, Brown University, Providence, RI                            | <i>Aug 2024</i> |
| Mixed Integer Programming Workshop (MIP 2024), Lexington, KY       | <i>Jun 2024</i> |
| Workshop on Combinatorial Optimization, Aussois, France            | <i>Jan 2024</i> |
| Special Seminar, IEOR, UC Berkeley                                 | <i>Nov 2023</i> |
| INFORMS 2023, Phoenix, AZ                                          | <i>Oct 2023</i> |
| Semester Program on Fast Algorithms, Simons Institute, UC Berkeley | <i>Sep 2023</i> |

*“Breaking the quadratic gap for strongly polynomial solvers to combinatorial linear programs”*

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|----------------------------------------------|-----------------|
| INFORMS 2024, Seattle, WA                    | <i>Oct 2024</i> |
| ISMP 2024, Montreal, Canada                  | <i>Jul 2024</i> |
| ICIAM 2023, Tokyo, Japan                     | <i>Aug 2023</i> |
| CS Theory Seminar, Columbia University       | <i>Apr 2023</i> |
| CGO Seminar, London School of Economics      | <i>Dec 2022</i> |
| ACO Seminar, Georgia Institute of Technology | <i>Nov 2022</i> |

*“Recovering strong polynomiality via straight-line-complexity”*

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| Semester Program on Discrete Optimization, ICERM, Brown University | <i>May 2023</i> |
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*“Linear Interior Point Methods”*

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| INFORMS Invited Optimization Society Talk, Atlanta, GA | <i>Mar 2026</i> |
| Korte Workshop, University of Bonn, Germany            | <i>Nov 2025</i> |

*“Trust Region Interior Point Methods”*

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| Discrete Optimization and Machine Learning, RIMS, Kyoto, Japan | <i>May 2025</i> |
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*“Interior point methods are not worse than Simplex”*

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| FOCS 2022, Denver, CO                                   | <i>Nov 2022</i> |
| Young Researchers Workshop, Cornell University          | <i>Oct 2022</i> |
| Workshop on Combinatorial Optimization, Cargese, France | <i>Sep 2022</i> |

*“A Faster Interior-Point Method for Sum-of-Squares Optimization”*

ICALP 2022, Paris, France

Jul 2022

*“Circuit imbalance measures and linear programming”*

Focused Workshop on Integer Programming, Rényi Center, Budapest

Aug 2023

Workshop on Circuits and Optimization, University of Colorado Denver

May 2023

Semester Program Seminar, ICERM, Brown University

Feb 2023

Theory Seminar, University of Washington

Mar 2022

Northwestern Junior Theorist Workshop, Northwestern University

Dec 2021

Hausdorff Trimester on Discrete Optimization, University of Bonn

Oct 2021

*“A scaling-invariant algorithm for linear programming whose running time only depends on the constraint matrix”*

SMAI MODE, Université Paris-Saclay

Sep 2020

Research Seminar, Columbia University

Sep 2020

STOC 2020, Chicago, IL (virtual)

Jun 2020

Seminar on Combinatorics, Games and Optimisation, London School of Economics

Dec 2019

*“Revisiting Tardos’s Framework for Linear Programming”*

HALG 2021, London School of Economics

Jun 2021

FOCS 2020, Duke University (virtual)

Nov 2020

Postgraduate Combinatorial Conference, Oxford University

Jun 2019

## TEACHING

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*Columbia University* ♦ *Course Instructor*

Advanced Optimization (IEOR 3609)

Spring 2025, 2026

Convex Optimization (EEOR 6616)

Spring 2026

Fast Algorithms for Linear Programs (IEOR 8100)

Spring 2025

*London School of Economics* ♦ *Teaching Assistant*

Reinforcement Learning — TA & joint course development

2022

Foundations of Machine Learning — TA & joint course development

2021, 2022

Algorithms and Programming

2020

Programming in C++

2019

Operational Research Methods

2019

Further Mathematical Methods

2018

*University of Bonn* ♦ *Teaching Assistant*

Combinatorics, Graphs and Matroids  
Calculus

2016  
2013

## SERVICE

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### Editorial Board

*since 2026*

Discrete Optimization.

### Reviewing

Conferences: ESA ♦ FOCS ♦ ICALP ♦ IPCO ♦ ISAAC ♦ SODA ♦ STOC.

Journals: *Discrete Optimization* ♦ *Journal of the ACM* ♦ *Mathematical Programming* ♦ *Mathematics of Operations Research* ♦ *Networks* ♦ *Proceedings of the AMS* ♦ *SIAM Journal on Discrete Mathematics* ♦ *SIAM Journal on Optimization*.

Grants: European Research Council (ERC).

Reviewer for *Mathematical Reviews* (MathSciNet).

### Program Committees

SODA 2025.

### Seminar Organization

*since Spring 2025*

Co-organizer, IEOR-DRO Seminar, Columbia University.

### PhD Admissions Committee

2025, 2026

IEOR Department, Columbia University.

### High School Outreach

2025

Guest lectures at Columbia Secondary School.

### Professional Memberships

INFORMS ♦ INFORMS Optimization Society ♦ MOS Mixed Integer Programming Society.